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| <b>LAB MODULE #7: AXIAL FLUX PLOT LAB</b> |
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**LABORATORY OBJECTIVES:**

1. Measure operational I-vs-V curve for miniature in-core (MIC) fission chamber detector and determine operational voltage plateau.
2. Verify linearity of detection channel response relative to neutron flux.
3. Measure the axial neutron flux profile in a reactor fuel assembly cooling channel and calculate the power peaking factor.
4. Observe shadowing of ex-core instrumentation channels through banking control rods.

**EQUIPMENT:**

- Miniature In-Core Fission Chamber detector – *See Figure #1.*
- Dry Instrument well – *See Figure #2.*
- Micro-Ammeter and HV Power Supply – *See Figure #3.*
- PULSTAR Reactor

**PRECONDITIONS:**

1. Reactor Daily Startup Checklist completed and signed by Designated Senior Reactor Operator (DSRO).
2. Reactor startup has been authorized by DSRO.
3. Licensed Reactor Operator is on duty at the reactor control console.
4. Instrument well inserted into fuel assembly adjacent to control rod.

**AXIAL FLUX PLOT PROCEDURE:**

1. Insert the MIC into the instrumentation well located in a fuel assembly cooling channel and connect to test electronics according to the attached diagram.
2. Bring reactor critical at 500W.
3. Run Current –vs- Voltage curve for MIC and record data on *Table #1*:

**Table #1: MIC I-V Curve Data**

| Voltage (V) | Current (A) |  | Voltage (V) | Current (A) |
|-------------|-------------|--|-------------|-------------|
| +10         |             |  | +120        |             |
| +20         |             |  | +140        |             |
| +40         |             |  | +160        |             |
| +60         |             |  | +180        |             |
| +80         |             |  | +200        |             |
| +100        |             |  | (+100)      |             |

4. Plot I-vs-V and determine plateau voltage. Set HV supply to provide this voltage for the test.
5. Measure linearity of detector channel to changes in power/flux by completing Table #2.

**Table #2: Linearity Test**

| Reactor<br>Power (Watts) | MIC Ammeter<br>Current (A) | Current Scaling<br>Factor |
|--------------------------|----------------------------|---------------------------|
| 500W                     |                            |                           |
| 1000W                    |                            |                           |
| 1500W                    |                            |                           |
| 2000W                    |                            |                           |

6. Verify linearity of MIC response by checking scaling factors between levels.
7. Vary reactor power to obtain an on-scale reading for the ammeter with MIC fully inserted in instrumentation wand.
8. Measure the cooling channel axial flux profile by withdrawing the MIC at a fixed rate and plotting the voltage –vs- time curve using the data acquisition system – *see Figure #4*.
9. The nominal sensitivity of the MIC is  $\sim 1.5 \times 10^{-17}$  Amps/neutrons/cm<sup>2</sup>/sec (or A/nv). This may be utilized along with the axial profile data to determine the neutron flux as a function of height in the reactor core.

Figure #1 – Typical In-Core Fission Chamber Detector Configurations:

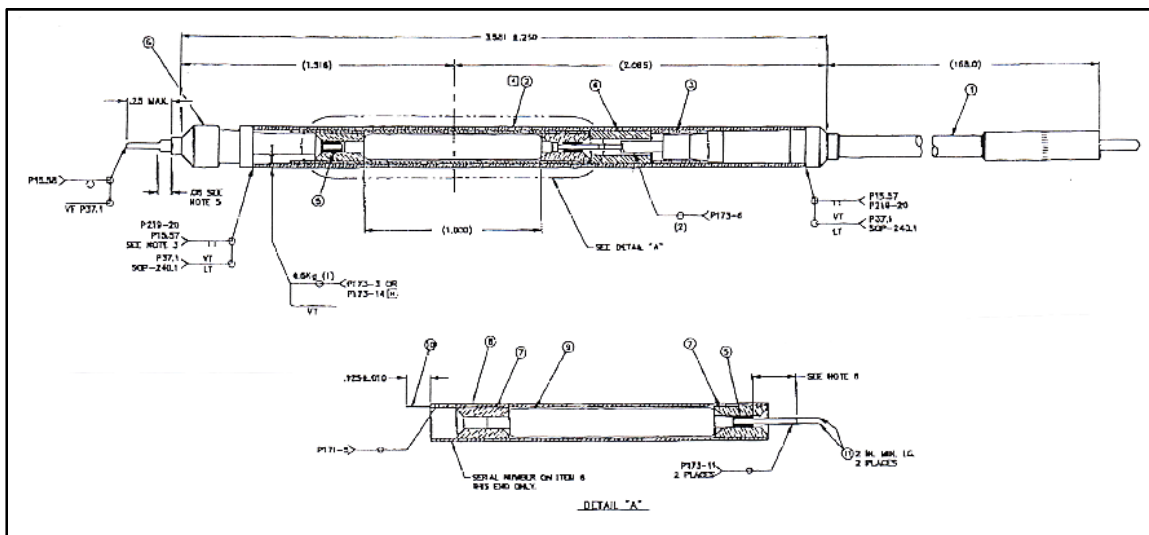


Figure #2 – Dry Instrument Well:

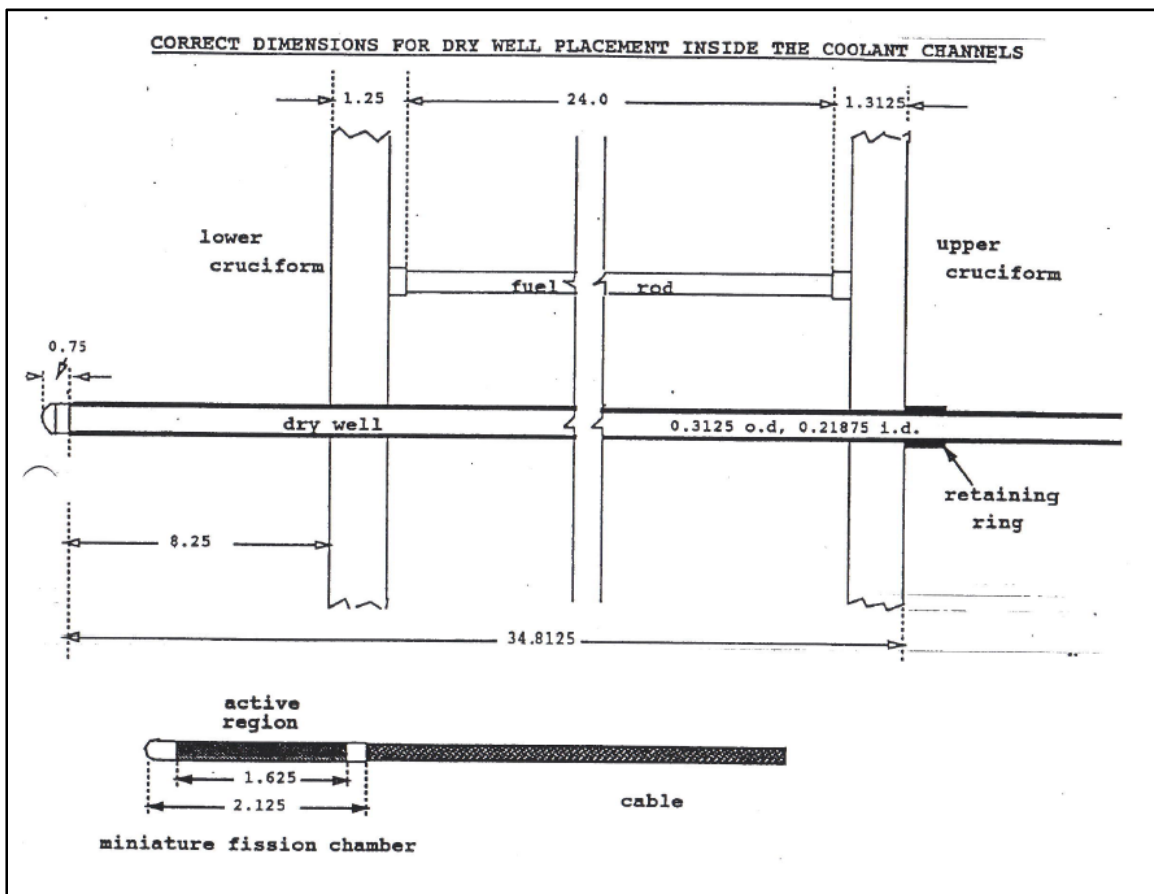


Figure #3 – Test Electronics Schematic

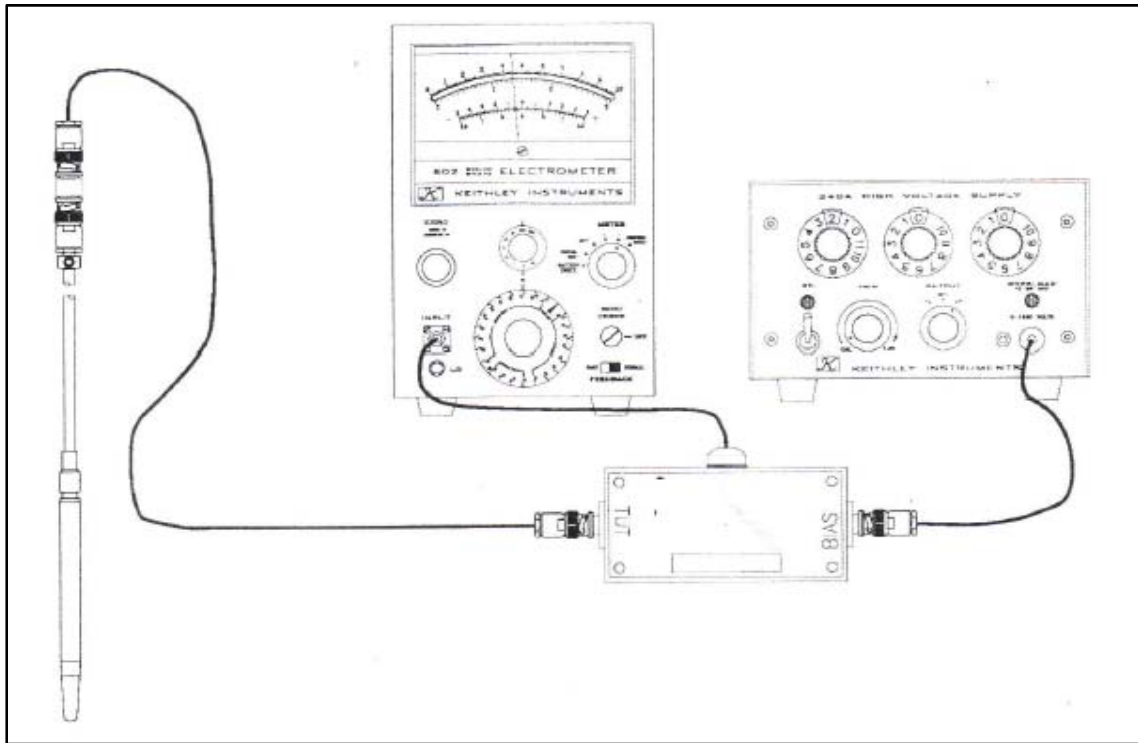


Figure #4 – Example Axial Flux Profile:

